

BDM — Topics from Live Sessions

Comprehensive Notes for End-Term Exam Preparation

Strategy · Analytics · Operations · Metrics

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1. Porter's Five Forces

Porter's Five Forces is a strategic framework that identifies five competitive forces shaping every industry. It helps determine industry structure and guides corporate strategy.

Competitive Rivalry

Intensity of competition among existing players.

- Many competitors or slow growth → high rivalry
- Low switching cost → customers easily switch
- High rivalry compresses margins and forces price competition

Threat of New Entrants

Risk that new players enter the market and erode market share.

- Low barriers (capital, brand, regulation) → high threat
- Incumbents respond with pricing or loyalty programmes
- Example: Easy to start an e-commerce reseller; hard to start an airline

Bargaining Power of Suppliers

Ability of suppliers to dictate price or terms.

- Few suppliers or unique inputs → high power
- Supplier can raise cost or reduce quality
- Example: ACE Gears depends on blank suppliers; fewer suppliers = higher risk

Bargaining Power of Buyers

Ability of customers to demand lower prices or better quality.

- Few buyers or large orders → high buyer power
- Commoditised products → easy to switch supplier
- Example: OEM buying gears in bulk has high bargaining power over ACE Gears

Threat of Substitutes

Risk of alternative products fulfilling the same need.

- Low switching cost → high threat
- Forces firms to innovate or cut price
- Example: BS4 → BS6 engine transition is a substitute threat for old gear assemblies

Exam Tip

- Know all 5 forces by name + what makes each force HIGH or LOW.
- Link each force to a business example (FabMart or ACE Gears).
- Higher force strength = lower industry profitability.

2. Industry Life Cycle

Every industry passes through four distinct stages. Understanding the stage determines pricing strategy, investment levels, and competitive approach.

Introduction

Product is new; few competitors; high R&D; cost; low volume.

- Strategy: Market skimming or penetration pricing
- Focus: Build awareness; capture early adopters
- Example: BS6 Gear Assemblies (GA9, GA10) entering market

Growth

Rapid revenue rise; new entrants; investment peaks.

- Strategy: Expand distribution; build brand
- Focus: Grab market share before market matures
- Prices begin to fall as competition increases

Maturity

Growth slows; competition intense; margins squeeze.

- Strategy: Cost leadership or differentiation
- Focus: Efficiency, loyalty, product variants
- Example: Established FMCG products in FabMart

Decline

Demand falls; players exit; consolidation occurs.

- Strategy: Harvest (milk remaining profit) or Exit
 - Example: BS4 Gear Assemblies — government mandate ended demand
 - Inventory of obsolete parts must be scrapped
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3. PESTEL Analysis

PESTEL is a macro-environment framework. It audits the external factors that affect an entire industry or business. Use it before strategy planning.

Factor	Meaning & Business Example
Political	Government policies, regulations, political stability. Example: BS4→BS6 mandate by GoI — forced product discontinuation at ACE Gears.
Economic	GDP, inflation, interest rates, consumer income. Example: COVID lockdown caused revenue crash visible in ACE Gears FY21 data.
Social	Demographics, culture, lifestyle changes. Example: Urban households shift spending from food to discretionary goods (NSS data).
Technological	Innovation, automation, digital platforms. Example: ERP systems, e-commerce platforms (FabMart), UPI payment rails.
Environmental	Climate, sustainability regulations. Example: BS6 fuel has lower sulphur (10 ppm vs 50 ppm in BS4) — environmental driver.
Legal	Labour laws, IP rights, consumer protection. Example: Anti-competitive merger rules use HHI to block monopolistic consolidation.

4. SWOT Analysis

SWOT is an internal + external audit tool. Strengths & Weaknesses are internal; Opportunities & Threats are external.

INTERNAL	EXTERNAL
Strengths Positive internal factors • Brand, capabilities, cost advantage • ACE Gears: established OEM relationships • FabMart: two-tier DC network	Opportunities Positive external factors • Growing markets, new tech, regulation changes • ACE Gears: BS6 transition = new product lines • FabMart: expanding e-commerce adoption
Weaknesses Negative internal factors • High cost, skill gaps, old tech • ACE Gears: labour shortage (COVID) • FabMart: stockouts on fast-moving items	Threats Negative external factors • Competitors, regulation, substitutes • ACE Gears: BS4 ban eliminated revenue • FabMart: delivery SLA competition

5. TOWS Matrix

TOWS is the action-oriented extension of SWOT. It combines internal and external factors to derive four strategic directions.

SO — Maxi-Maxi

- Use Strengths to exploit Opportunities
 - Example: ACE Gears uses manufacturing expertise (S) to capture BS6 demand (O)

WO — Mini-Maxi

- Overcome Weaknesses by exploiting Opportunities
 - Example: FabMart invests in DC automation (O) to fix stockout weakness (W)

ST — Maxi-Mini

- Use Strengths to counter Threats
 - Example: ACE Gears uses OEM relationships (S) to survive new Chinese competitors (T)

WT — Mini-Mini

- Minimise Weaknesses and avoid Threats
 - Example: Phasing out BS4 production (W+T) to cut losses

Difference: SWOT vs TOWS

- SWOT = audit tool (what is the situation?)
- TOWS = strategy tool (what should we DO about it?)
- TOWS uses SWOT as input and generates 4 strategic options.

6. BCG Matrix

The BCG (Boston Consulting Group) Growth-Share Matrix classifies a company's products/SBUs based on market growth rate (Y-axis) and relative market share (X-axis) to guide resource allocation.

Stars

High Growth, High Market Share

- Market leaders in fast-growing segments
- Need heavy investment to maintain position
- Eventually become Cash Cows as growth slows
- ACE Gears: New BS6 Gear Assemblies (GA9, GA10) if growth is high

Cash Cows

Low Growth, High Market Share

- Mature products generating steady cash
- Require little investment; fund other units
- Strategy: Milk for profit, defend position
- FabMart: Top-revenue SKUs like M01, M07 — high revenue, stable

Question Marks (Problem Children)

High Growth, Low Market Share

- Uncertain potential; need investment or divestment
- Strategy: Invest heavily (→ Star) or Drop (→ Dog)
- ACE Gears: GA9 newly launched — low volume but growing market

Dogs

Low Growth, Low Market Share

- Weak competitors in stagnant markets
- Strategy: Divest or harvest remaining value
- ACE Gears: BS4 Gear Assemblies (GA1, GA2) — mandated phase-out

Exam Tip

- Axes: Y = Market Growth Rate, X = Relative Market Share
- Stars need cash IN; Cash Cows generate cash OUT.
- Link BCG quadrants to Industry Life Cycle stages: Intro→Question Mark, Growth→Star, Maturity→Cash Cow, Decline→Dog.

7. ABC Analysis & Pareto Principle

Pareto Principle (80/20 Rule)

Roughly 80% of consequences come from 20% of causes.

- Revenue Pareto: Top 20% SKUs → 80% revenue (FabMart: top 6 of 30 SKUs = 80% revenue ✓)
- Volume Pareto: top 36% SKUs → 80% volume (FabMart did NOT follow 80/20 strictly)
- Business use: Allocate resources to the vital few; monitor movement in Pareto rank.
- A shift in Pareto rank signals changing market sentiment — act immediately.

ABC Analysis

Classifies inventory into three tiers based on value and control needed.

Category	% Items	% Value	Control Level
A Items	10–20% of SKUs	70–80% of value	Tight control, accurate records, frequent review
B Items	20–30% of SKUs	15–20% of value	Moderate control, regular monitoring
C Items	50–70% of SKUs	5–10% of value	Minimal control, simple records, bulk orders

- Head-Torso-Tail = another name for A-B-C (used in FabMart case)
- Gear blanks at ACE Gears are Category B — important but not highest value
- Even tail (C) items must be stocked to avoid losing niche customers
- ABC × DOI: A items need lower DOI (fast replenishment); C items can hold more

Warehouse Positioning Logic (FabMart)

- Fast movers (high volume) → near dock/processing bay for speed
- High value items → secure storage (lock and key), even if far from bay
- High volume + High value → security wins over speed

8. Lead, Lag & Surrogate Indicators

Lead Indicators

Forward-looking / predictive metrics — measured BEFORE the outcome occurs.

- Help managers take proactive action
- Examples: number of sales calls made, safety stock level, order pipeline, daily order rate
- FabMart: tracking DOI daily is a lead indicator to prevent future stockout
- ACE Gears: monitoring blank inventory before production gaps occur

Lag Indicators

Outcome / result metrics — measured AFTER the outcome occurs.

- Confirm whether goals were achieved
- Examples: quarterly revenue, scrap percentage, OEE score, monthly sales volume
- ACE Gears: end-of-month actual production vs plan is a lag indicator
- Cannot be changed for the current period — only inform future decisions

Surrogate Indicators

Proxy metrics used when direct measurement is unavailable or costly.

- Indirect evidence of a true variable
- Examples: CIBIL score as surrogate for repayment ability; ZIP code as proxy for income
- Consumer Pyramid survey data as surrogate for national consumption patterns
- Caution: surrogate may introduce bias — location-based credit decision is discrimination

Type	When measured	Action possible?	Example
Lead	Before outcome	Yes — proactive	Low DOI → reorder now
Lag	After outcome	No — reactive	Revenue fell 10% last Q
Surrogate	Indirect proxy	Depends	CIBIL → creditworthiness

9. Market Concepts

Market Share

The percentage of total industry sales held by a firm.

- Market Share = (Firm Revenue / Total Industry Revenue) × 100
- Used in HHI calculation: $HHI = \sum (\text{market share})^2$ for all firms
- HHI < 1000: competitive; 1000–1800: moderate; > 1800: concentrated/monopolistic
- Anti-competitive mergers are blocked using HHI thresholds

Market Growth Rate

The annual percentage increase in total market size.

- Y-axis of BCG matrix
- High growth = attractive market but requires investment
- Formula: Growth Rate = ((Current Year Sales – Previous Year Sales) / Previous Year Sales) × 100
- ACE Gears: compare FY20 vs FY22 (skip FY21 — COVID anomaly)

Market Segmentation

Dividing the total market into distinct sub-groups with similar needs or behaviours.

- Geographic: North/East/West/South (ACE Gears regional analysis)
 - Demographic: income strata (NSS consumption baskets at 15k / 30k / 50k household income)
 - Behavioural: usage frequency, brand loyalty (FabMart FMCG vs Mobiles)
 - Psychographic: lifestyle, values
 - Fintech: segment by payment instrument (BNPL vs CC vs DC users)
 - Top-down segmentation: estimate total market first, then drill into segment share
 - Bottom-up segmentation: aggregate individual data to build market picture
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10. OEE — Overall Equipment Effectiveness

OEE is the gold standard metric for manufacturing productivity. It measures the percentage of manufacturing time that is truly productive.

OEE Formula

OEE = Availability (A) × Performance (P) × Quality (Q)

A = (Planned Shifts – Breakdown Shifts) / Planned Shifts

P = Actual Output / Planned Output (Hobbing)

Q = Good Output (Broaching) / Total Input (Hobbing output — excl. last shift)

Components Explained

Availability

- Percentage of scheduled time the machine was actually running.
 - Note: Breakdowns, emergency maintenance reduce A. Maintenance crew competence drives A.

Performance

- Ratio of actual speed to designed/planned speed.
 - Note: Slow machines reduce P. Measured as actual output vs planned output per shift.

Quality

- Ratio of good (non-scrap) output to total output.
 - Note: Defective/scrap parts reduce Q. Shift worker and material quality drive Q.

Key Nuances for Exam

- Quality denominator = Hobbing output EXCLUDING the last shift (last hobbing output is not fed into broaching in the same week)
- Non-operational shifts (changeover, maintenance) are EXCLUDED from Performance and Quality calculation
- Scrap = Hobbing output – Broaching output (assuming no broaching scrap)
- Scrap % = Scrap / Hobbing output × 100 — expect lower than planned 2.5% in the case study
- Shift 1 consistently produces more scrap than Shift 2 (real-world observation from ACE Gears data)
- Material Cost of Scrap = Number of scrap pieces × Cost of blank

OEE = 100% means

- 100% Quality: only good parts produced
- 100% Performance: running at maximum designed speed
- 100% Availability: zero downtime — no stop time

11. EOQ & Inventory Concepts

Key Inventory Terms

Term	Definition
Opening Stock	Inventory at the start of the period
Closing Stock	Opening Stock – Sales + Transfers IN (floor at 0, cannot go negative)
Days of Inventory (DOI)	Opening Stock / Average Daily Sales — directional signal, not precise
Safety Stock	Buffer stock held against demand or supply uncertainty
Reorder Point (ROP)	Minimum inventory level that triggers a new order
Lead Time	Time between placing an order and receiving it (ACE Gears: 1 month)
Lead Time Demand	Expected demand during the lead time period
EOQ	Optimal order quantity minimising total holding + ordering cost
MOQ	Minimum Order Quantity — set by the supplier, not the buyer
Stockout	Item out of stock — causes lost sales and customer dissatisfaction

Key Formulas

Closing Stock = Opening Stock – Sales + Transfer In (min value = 0)

Safety Stock = Maximum Demand – Average Demand

Reorder Point (ROP) = Lead Time Demand + Safety Stock

EOQ = $\sqrt{2 \times D \times S / H}$ where D = Annual Demand, S = Order Cost, H = Holding Cost/unit/yr

DOI = Stock on Hand / Average Daily Sales

Business Insights

- DOI ignores Transfer In — it is a directional warning signal only
- Closing stock floors at zero: even if formula gives negative, record as 0 (Mother DC fills the gap)
- Oscillations in blank inventory at ACE Gears are caused by lead time mismatch with production
- MOQ from supplier quantises order quantities — cannot order exactly EOQ amount
- Obsolete BS4 blanks should be sold as scrap metal (sunk cost, no future value)
- If DOI < 7 days → raise alarm and expedite replenishment (FabMart rule)
- If DOI < 2 days at child DC → ship from mother DC immediately

12. MAPE — Forecasting Accuracy

Mean Absolute Percentage Error (MAPE) measures the accuracy of a forecast. Lower MAPE = better forecast.

Formula

$$\text{MAPE} = (1/n) \times \sum |(\text{Actual} - \text{Forecast}) / \text{Actual}| \times 100$$

Key Rules

- Non-operational shifts (maintenance, changeover) are EXCLUDED from MAPE calculation
 - MAPE is expressed as a percentage — 0% = perfect forecast
 - High MAPE means poor planning accuracy → need better demand forecasting
 - Used in ACE Gears to measure how well production plan matched actual output
 - In FabMart: trend analysis over 15 days is insufficient to calculate meaningful MAPE — need longer time series
 - COVID year (FY21) is excluded from YoY comparisons because it is an outlier
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13. Revenue & Margin Analysis

Revenue Calculation

$$\text{Revenue} = \text{Quantity Sold} \times \text{Selling Price}$$

- In FabMart: Revenue by SKU = VLOOKUP price into Sales Data, then multiply by quantity
- In ACE Gears: Revenue per GA per month = Qty × Price from Sales Details sheet

Cost Structure

- **Material Cost:** Cost of raw material (blanks) per GA. Looked up from blank cost table × blank-gear relation.
- **Labour Cost:** Total wage of production line workers ÷ number of units produced.
- **Production Overhead:** Maintenance, electricity, supervisor cost — apportioned by volume share.
- **G&A; Overhead:** Sales team, logistics, HQ costs. NOT included in COGS for Gross Margin calculation.

Key Margin Formulas

$$\text{COGS} = \text{Material Cost} + \text{Labour Cost} + \text{Production Overhead (NOT G\&A;)}$$

$$\text{Gross Margin} = \text{Selling Price} - \text{COGS}$$

$$\text{Gross Margin \%} = (\text{Selling Price} - \text{COGS}) / \text{Selling Price} \times 100 \leftarrow \text{denominator is REVENUE, not cost}$$

$$\text{Net Margin \%} = (\text{Revenue} - \text{COGS} - \text{G\&A;}) / \text{Revenue} \times 100$$

Critical Nuances

- G&A; overhead is EXCLUDED from COGS when computing Gross Margin (only subtracted at Net Margin level)
- GM% uses Revenue (Selling Price) as denominator — common exam mistake is to use Cost
- A high GM% = company covers production costs efficiently and retains more from each rupee of revenue
- What-if Analysis (Goal Seek in Excel): find the price needed to achieve a target GM%
- In Tier-1 manufacturing (ACE Gears), OEM sets the price — you CANNOT increase selling price; you must cut COGS
- Trendline in scatter plot is more analytically useful than a 2×2 grid for portfolio analysis
- GA7: overpriced relative to volume — above trendline
- GA6: underpriced volume trap — below trendline, high volume but low revenue
- GA9: new BS6 product, low volume is expected, NOT a failing product

14. Quick Revision Cheatsheet

FRAMEWORKS AT A GLANCE

- Porter's 5 Forces: Rivalry | New Entrants | Suppliers | Buyers | Substitutes
- Industry Life Cycle: Introduction → Growth → Maturity → Decline
- PESTEL: Political | Economic | Social | Technological | Environmental | Legal
- SWOT: Strengths + Weaknesses (internal) | Opportunities + Threats (external)
- TOWS: SO (use S for O) | WO (fix W via O) | ST (S vs T) | WT (minimise both)
- BCG: Stars (HG/HS) | Cash Cows (LG/HS) | Question Marks (HG/LS) | Dogs (LG/LS)

INVENTORY FORMULAS

- Closing Stock = Opening – Sales + Transfer In (min 0)
- DOI = Stock / Avg Daily Sales
- Safety Stock = Max Demand – Avg Demand
- ROP = Lead Time Demand + Safety Stock
- EOQ = $\sqrt{2DS/H}$

MANUFACTURING FORMULAS

- OEE = Availability × Performance × Quality
- A = (Planned – Breakdown Shifts) / Planned Shifts
- P = Actual Output / Planned Output
- Q = Good Output / Total Hobbing Output (excl. last shift)
- Scrap = Hobbing output – Broaching output
- MAPE = $(1/n) \sum |Actual - Forecast| / Actual \times 100$ (exclude non-operational shifts)

MARGIN FORMULAS

- Revenue = Qty × Price
- COGS = Material + Labour + Production Overhead (NO G&A;)
- Gross Margin = Selling Price – COGS
- Gross Margin % = $(SP - COGS) / SP \times 100$ (Revenue is denominator!)
- Net Margin = Revenue – COGS – G&A;

INDICATORS

- Lead: predictive, proactive (DOI, pipeline, safety stock level)
- Lag: outcome, reactive (revenue, OEE score, scrap %)
- Surrogate: proxy when direct measure unavailable (CIBIL for repayment ability)

PARETO & ABC

- 80/20 Rule: 80% revenue from 20% SKUs
- A items: 10–20% SKUs, 70–80% value — tight control
- B items: 20–30% SKUs, 15–20% value — moderate control
- C items: majority SKUs, 5–10% value — loose control

- Warehouse: Fast movers near dock; High value in secure bay

MARKET CONCEPTS

- Market Share = $(\text{Firm Sales} / \text{Total Market}) \times 100$
- HHI = $\sum (\text{market share})^2$ | <1000: competitive, >1800: concentrated
- Segmentation: Geographic | Demographic | Behavioural | Psychographic
- Growth Rate = $(\text{Current} - \text{Previous}) / \text{Previous} \times 100$
- BCG Y-axis = Growth Rate | BCG X-axis = Relative Market Share

EXAM WATCHOUTS

- GM% denominator = Revenue (NOT cost price)
- G&A; is excluded from COGS (only at net margin level)
- OEE Quality excludes LAST SHIFT of hobbing output
- DOI ignores Transfer In — directional only
- Closing stock cannot go negative — floor at zero
- India fiscal year: Apr–Mar | Jan–Mar = Q4 (not Q1)
- Trendline > 2×2 grid for portfolio scatter analysis
- COVID year FY21 is anomaly — compare FY22 vs FY20
- SWOT = What is situation? | TOWS = What to DO?
- Stars need cash IN; Cash Cows generate cash OUT

End of Notes | BDM Live Session Topics | Good luck on your exam!